Math 542

Homework 1

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Exercise 1. *Show that is* A *is a deformation retract of* X*, and* B *is a deformation retract of* A*, then* B *is a deformation retract of* X*.*

Proof. Since A is a deformation retract of X, there exists a continuous function $F : X \times I \to X$ such that:

$$\begin{aligned} F(x,0) &= x \\ F(x,1) &= r_A(x) \\ F(a,t) &= a \text{ for all } t \in I, \end{aligned}$$

where $r_A : X \to A$ is a retraction. Since B is a deformation retract of A, there exists a continuous function $G : A \times I \to A$ such that:

$$\begin{aligned} G(a,0) &= a \\ G(a,1) &= r_B(a) \\ G(b,t) &= b \text{ for all } t \in I, \end{aligned}$$

where $r_B : A \to B$ is a retraction. Now define $H : X \times I \to X$ by:

$$H(x,t) := \begin{cases} F(x,2t), & 0 \leqslant t \leqslant \frac{1}{2} \\ G(F(x,1),2t-1), & \frac{1}{2} \leqslant t \leqslant 1 \end{cases}.$$

Then: Woohoo! It's not broken!

$$\begin{aligned} &H(x,0) = F(x,0) = x, \\ &H(x,1) = G(F(x,1),1) = G(r_A(x),1) = (r_B \circ r_A)(x), \\ &H(b,t) = F(b,2t) = b \text{ for } 0 \leqslant t \leqslant \frac{1}{2}, \\ &H(b,t) = G(F(b,t),2t-1) = G(b,2t-1) = b \text{ for } \frac{1}{2} \leqslant t \leqslant 1. \end{aligned}$$

Clearly $(r_B \circ r_A) : X \to B$ is a retraction, which establishes the proof. □

Exercise 2.

(a) *Infinite cyclic.*

(b) *Figure eight.*

(c) *Infinite cyclic.*

(d) *Infinite cyclic.*

(e) *Trivial.*

(f) *Infinite cyclic.*

(g) *Infinite cyclic.*

(h) *Trivial.*

(i) *Infinite cyclic.*

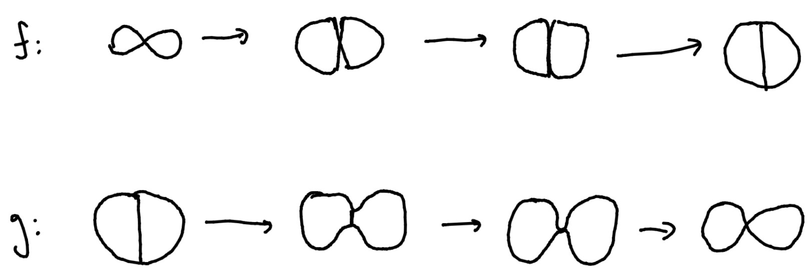
(j) *Infinite cyclic.*

(k) *Figure eight.*

(l) *Trivial.*

Exercise 4. *Let* X *be the figure eight and let* Y *be the theta space. Describe maps* $f : X \longrightarrow Y$ *and* $g : Y \longrightarrow X$ *that are homotopy inverses of each other.*

Proof. The following maps are described in the drawing below.



$\square$

Exercise 5. *Show that* X *is contractible if and only if* X *has the homotopy type of a one-point space.*

Proof. Suppose X is contractible. Then id_X is homotopic to a constant map. Let $c : X \longrightarrow X$ be such a constant map given by $c(x) = x_0$ for all $x \in X$. Define $f : X \longrightarrow \{x_0\}$ by $f(x) = c(x)$. Define $g : \{x_0\} \longrightarrow X$ by $g(x) = \mathrm{id}_X(x)$. Then $g \circ f = c \simeq \mathrm{id}_X$ by hypothesis, and $f \circ g = \mathrm{id}_{\{x_0\}}$ (which is clearly stronger than being homotopic to $\mathrm{id}_{\{x_0\}}$). Thus X and $\{x_0\}$ are homotopy equivalent.

Conversely, let X be homotopy equivalent to $\{x_0\}$. There exists continuous functions $f : X \longrightarrow \{x_0\}$ and $g : \{x_0\} \longrightarrow X$ such that $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_{\{x_0\}}$. But note that $f \circ g$ is a constant map. Hence X is contractible. $\square$

Exercise 6. *Show that a retract of a contractible space is contractible.*

Proof. Let X be contractible. Let $A \subset X$ be a retraction. Then there exists $r : X \to A$ such that $r|_A = \mathrm{id}_A$. Since X is contractible, there exists $F : X \times I \to X$ such that $F(x, 0) = \mathrm{id}_X(x)$ and $F(x, 1) = c(x)$, where $c : X \to X$ is a constant map. Define $G : A \times I \to A$ by $G(a, t) := r(F(a, t))$. Then $G(a, 0) = r(F(a, 0)) = r(\mathrm{id}_A(a)) = r(a) = a$, and $G(a, 1) = r(F(a, 1)) = r(c(a))$. Note that $r \circ c : A \to A$ is constant, hence A is contractible. □

Exercise 7. *Let A be a subspace of X; let $j : A \to X$ be the inclusion map, and let $f : X \to A$ be a continuous map. Suppose there is a homotopy $H : X \times I \to X$ between the map $j \circ f$ and the identity map X.*

(a) *Show that if H maps $A \times I$ into A, then j_* is an isomorphism.*

(b) *Give an example in which j_* is not an isomorphism.*

Proof. (a) Since $H(A \times I) \subseteq A$, the map $H|_{A \times I} : A \times I \to A$ satisfies the following:

$$H\big|_{A\times I}(a, 0) = (j \circ f)(a) = f(a) = (f \circ j)(a),$$
$$H\big|_{A\times I}(a, 1) = \mathrm{id}_X(a) = a = \mathrm{id}_A(a).$$

Hence $f \circ j \simeq \mathrm{id}_A$. Since $j \circ f = \mathrm{id}_X$ by assumption, this means A and X are homotopy equivalent. Thus $j_* : \pi_1(A, a) \to \pi_1(X, a)$ is an isometry for all $a \in A$.

(b) Let $f : B^2 \to S^1$ be a constant map. Then $j \circ f$ is constant, and since B^2 is contractible, this constant map is homotopic to id_{B^2}. But $j_* : \pi_1(S^1, a) \to \pi_1(B^2, a)$ is not an isomorphism, as $\pi_1(S^1, a) \cong \mathbb{Z}$ and $\pi_1(B^2, a)$ is trivial. □

Exercise 8. *Find a space X and a point x_0 of X such that the inclusion $\{x_0\} \to X$ is a homotopy equivalence, but $\{x_0\}$ is not a deformation retract of X.*

Proof. Let X be the comb space, and let $x_0 = (0, 1)$, the top of the limiting vertical segment. Note that the inclusion $\{x_0\} \to X$ is a homotopy equivalence, as each point of the comb space can be deformed to $(0, 1)$. Suppose towards contradiction $\{x_0\}$ is a deformation retract of X, that is, suppose there exists a continuous function $H : X \times I \to X$ that is a deformation retract onto $\{x_0\}$. Choose a small neighborhood U of x_0 that does not intersect the bottom horizontal segment of the space. Since $H(x_0, t) = x_0$ for all $t \in I$, we have $\{x_0\} \times I \subset H^{-1}(U)$. The set $H^{-1}(U)$ is open in $X \times I$, and I is compact. By the tube lemma, there is a neighborhood V of x_0 such that $V \times I \subset H^{-1}(U)$. This means that for each $y \in V$, the homotopy $t \mapsto H(y, t)$ stays inside U and joints y to x_0. Hence the path component of U containing x_0 must contain the whole neighborhood V. But this is a contradiction, as the comb space is not locally path connected about the point $(0, 1)$. Thus $\{x_0\}$ is not a deformation retract of X. □

We're about to test some of the snippers.

Theorem 0.0.1. *A space X is a contractible if and only if. I've noticed that using the inline math snippet, it sometimes adds a space. For example, $\epsilon > 0$ adds a space, and it's kinda weird!. Consider the function $f|_A$.*